

Week 11: Linear Regression

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Imports

```
# Load necessary libraries  
library(dplyr)
```

```
##  
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':  
##  
##     filter, lag
```

```
## The following objects are masked from 'package:base':  
##  
##     intersect, setdiff, setequal, union
```

```
library(GGally)
```

```
## Loading required package: ggplot2
```

```
## Registered S3 method overwritten by 'GGally':  
##   method from  
##   +.gg      ggplot2
```

```
library(lsa)
```

```
## Loading required package: SnowballC
```

```
library(tidyverse)
```

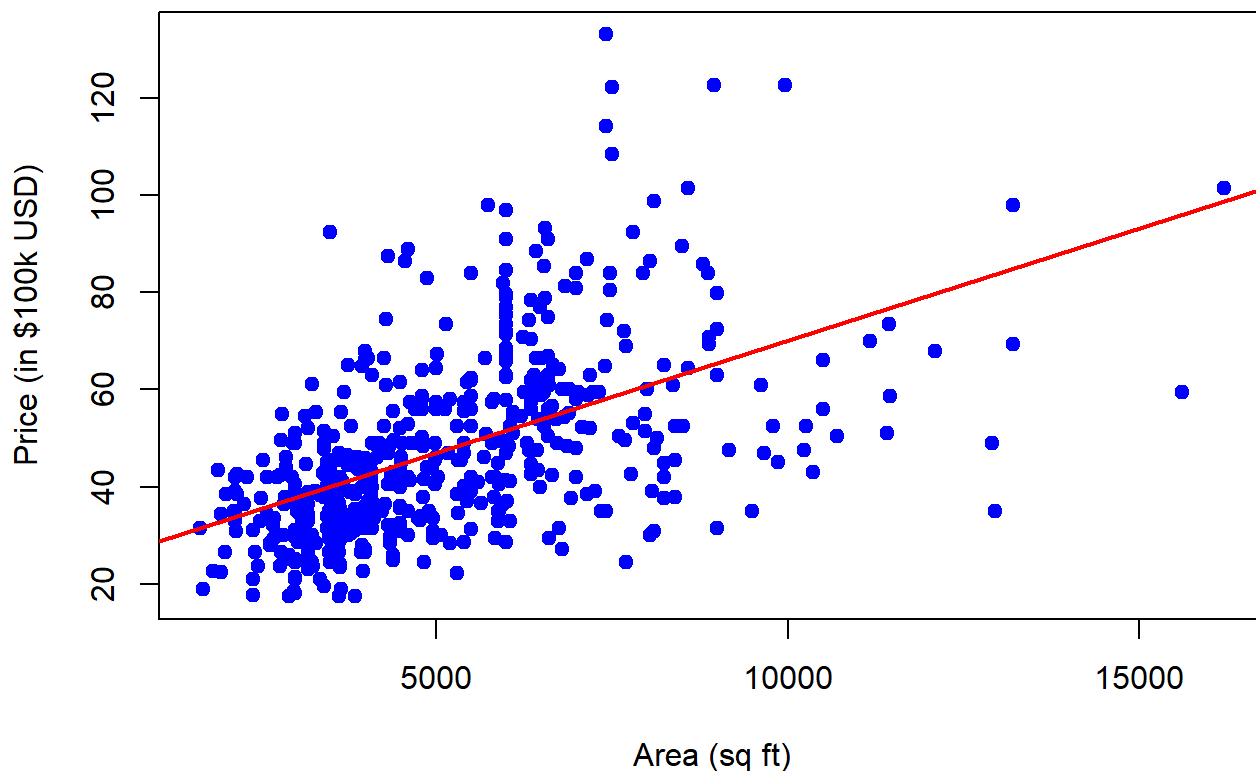
```
## — Attaching core tidyverse packages ————— tidyverse 2.0.0 —  
## ✓ forcats    1.0.0      ✓ stringr    1.5.1  
## ✓ lubridate  1.9.4      ✓ tibble     3.2.1  
## ✓ purrr      1.0.4      ✓ tidyr      1.3.1  
## ✓ readr      2.1.5
```

```
## — Conflicts ————— tidyverse_conflicts() —  
## * dplyr::filter() masks stats::filter()  
## * dplyr::lag() masks stats::lag()  
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts  
to become errors
```

Create a linear regression model where “area” infers price (i.e., area is used to predict price).

```
# Load housing data  
housing_df <- read.csv("housing.csv")  
  
# Convert price to $100k USD  
housing_df$price_usd_100k <- housing_df$price / 100000  
# Fit linear regression model  
lm_model <- lm(price_usd_100k ~ area, data = housing_df)  
# Plot: Area vs Price with regression line  
plot(  
  housing_df$area, housing_df$price_usd_100k,  
  main = "Housing Price vs Area (Linear Regression)",  
  xlab = "Area (sq ft)", ylab = "Price (in $100k USD)",  
  pch = 19, col = "blue"  
)  
  
abline(lm_model, col = "red", lwd = 2)
```

Housing Price vs Area (Linear Regression)



Get a summary of your first model and explain your results (e.g., R2, adj R2, RMSE, MSE).

```
# Display model summary  
summary(lm_model)
```

```
##
## Call:
## lm(formula = price_usd_100k ~ area, data = housing_df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -48.671 -10.222  -2.001   6.830  74.848
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.387e+01  1.745e+00  13.68  <2e-16 ***
## area         4.620e-03  3.123e-04  14.79  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 15.81 on 543 degrees of freedom
## Multiple R-squared:  0.2873, Adjusted R-squared:  0.286
## F-statistic: 218.9 on 1 and 543 DF,  p-value: < 2.2e-16
```

```
# Get values
r_squared <- summary(lm_model)$r.squared
adj_r_squared <- summary(lm_model)$adj.r.squared
rmse <- sqrt(mean(lm_model$residuals^2))
mse <- mean(lm_model$residuals^2)
# Print results
cat("R-squared:", r_squared, "\n")
```

```
## R-squared: 0.2872932
```

```
cat("Adjusted R-squared:", adj_r_squared, "\n")
```

```
## Adjusted R-squared: 0.2859806
```

```
cat("RMSE:", rmse, "\n")
```

```
## RMSE: 15.77613
```

```
cat("MSE:", mse, "\n")
```

```
## MSE: 248.8861
```

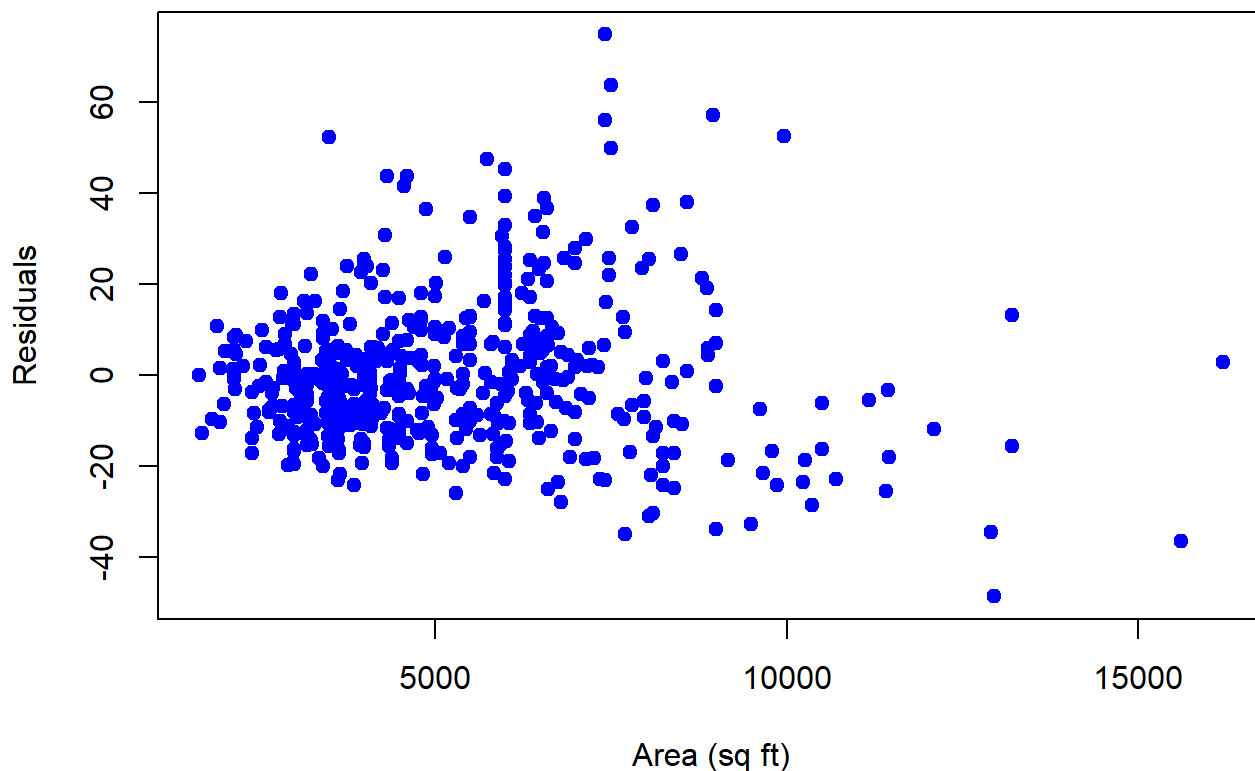
```
cat(
  "### Model Summary\n",
  "R-squared - 28.73% of the variation in the outcome is explained by the model.\n",
  "Adjusted R-squared - 28.59% close to R-squared, indicating added predictors may not be contributing much.\n",
  "RMSE - 15.77 The models predictions are off by about 15.78 units on average.\n",
  "MSE - 248.88 The average squared prediction error; useful for comparing models (lower is better).\n",
  "\nOverall, the model has limited explanatory power and large prediction error.\n"
)
```

```
## ### Model Summary
## R-squared - 28.73% of the variation in the outcome is explained by the model.
## Adjusted R-squared - 28.59% close to R-squared, indicating added predictors may not be contributing much.
## RMSE - 15.77 The models predictions are off by about 15.78 units on average.
## MSE - 248.88 The average squared prediction error; useful for comparing models (lower is better).
##
## Overall, the model has limited explanatory power and large prediction error.
```

Get the residuals of your model (you can use ‘resid’ or ‘residuals’ functions) and plot them. What does the plot tell you about your predictions?

```
# Get residuals
residuals_lm <- resid(lm_model)
# Plot residuals
plot(
  housing_df$area, residuals_lm,
  main = "Residuals of Linear Regression Model",
  xlab = "Area (sq ft)", ylab = "Residuals",
  pch = 19, col = "blue"
)
```

Residuals of Linear Regression Model



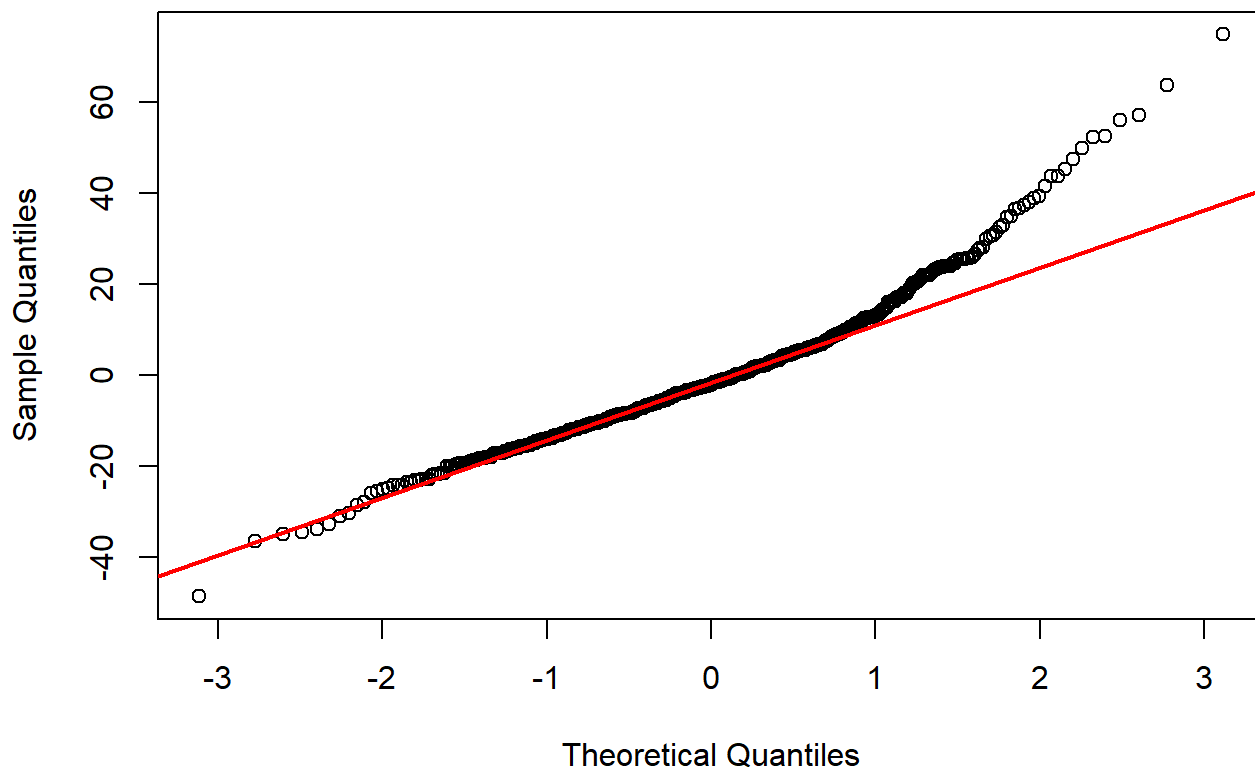
```
cat("The residuals show more variance as area increases. This suggests that the model's  
predictions are less accurate for larger areas, indicating potential non-linearity or the  
need for additional predictors.")
```

```
## The residuals show more variance as area increases. This suggests that the model's pr  
edictions are less accurate for larger areas, indicating potential non-linearity or the  
need for additional predictors.
```

Use a qq plot to observe your residuals. Do your residuals meet the normality assumption?

```
# Q-Q plot of residuals  
qqnorm(residuals_lm)  
qqline(residuals_lm, col = "red", lwd = 2)
```

Normal Q-Q Plot



```
cat("Both the end and beginnig of the Q-Q plot show that the residuals do not meet the n  
ormality assumption. The residuals deviate from the diagonal line.")
```

```
## Both the end and beginnig of the Q-Q plot show that the residuals do not meet the nor  
mality assumption. The residuals deviate from the diagonal line.
```

Create a linear regression model that uses multiple predictor variables to predict Sale Price (feel free to derive new predictors from existing ones). Explain why you think each of these variables may add explanatory value to the model.

```
# Derive new predictors
housing_df <- housing_df %>%
  mutate(
    total_rooms = bedrooms + bathrooms,
    space_per_story = ifelse(stories == 0, NA, area / stories) # Avoid division by zero
  )

# Fit multiple linear regression model, this time only using the total rooms and space per story
multi_lm_model <- lm(price_usd_100k ~ total_rooms + space_per_story, data = housing_df)

# Using this for later
all_variables_lm_model <- lm(price_usd_100k ~ total_rooms + space_per_story + area + bedrooms + bathrooms + stories, data = housing_df)

# Explain variables
cat(
  "New Predictors:\n",
  "- Total Rooms: Combines bedrooms and bathrooms for overall living capacity.\n",
  "- Space per Story: Multi story homes might be weighted differently given the average size.\n"
)
```

```
## New Predictors:
## - Total Rooms: Combines bedrooms and bathrooms for overall living capacity.
## - Space per Story: Multi story homes might be weighted differently given the average size.
```

Get a summary of this GLM and explain your results.

```
# Display model summary
summary(multi_lm_model)
```

```
##
## Call:
## lm(formula = price_usd_100k ~ total_rooms + space_per_story,
##     data = housing_df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -42.425 -10.443  -1.480   7.546  69.565
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -1.6817646   3.2735021  -0.514    0.608
## total_rooms    10.0916376   0.6596388  15.299 < 2e-16 ***
## space_per_story  0.0018463   0.0003044   6.065 2.48e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 15.58 on 542 degrees of freedom
## Multiple R-squared:  0.3089, Adjusted R-squared:  0.3063
## F-statistic: 121.1 on 2 and 542 DF,  p-value: < 2.2e-16
```

```
# Get values
r_squared <- summary(multi_lm_model)$r.squared
adj_r_squared <- summary(multi_lm_model)$adj.r.squared
rmse <- sqrt(mean(multi_lm_model$residuals^2))
mse <- mean(multi_lm_model$residuals^2)
# Print results
cat("R-squared:", r_squared, "\n") # 0.3088521
```

```
## R-squared: 0.3088521
```

```
cat("Adjusted R-squared:", adj_r_squared, "\n") # 0.3063018
```

```
## Adjusted R-squared: 0.3063018
```

```
cat("RMSE:", rmse, "\n") # 15.53568
```

```
## RMSE: 15.53568
```

```
cat("MSE:", mse, "\n") # 241.3575
```

```
## MSE: 241.3575
```

```
cat("R-squared -", round(r_squared, 3), ": About", round(r_squared * 100, 1), "% of the
variation in sale price is explained by the model.\n")
```

```
## R-squared - 0.309 : About 30.9 % of the variation in sale price is explained by the model.
```

```
cat("Adjusted R-squared -", round(adj_r_squared, 3), ": Adjusts R-squared for the number of predictors; still shows a moderate fit.\n")
```

```
## Adjusted R-squared - 0.306 : Adjusts R-squared for the number of predictors; still shows a moderate fit.
```

```
cat("RMSE -", round(rmse, 2), ": On average, predictions are off by about $", round(rmse * 100000, 0), ".\n")
```

```
## RMSE - 15.54 : On average, predictions are off by about $ 1553568 .
```

```
cat("MSE -", round(mse, 2), ": This is the average squared error; lower values indicate better model performance.\n")
```

```
## MSE - 241.36 : This is the average squared error; lower values indicate better model performance.
```

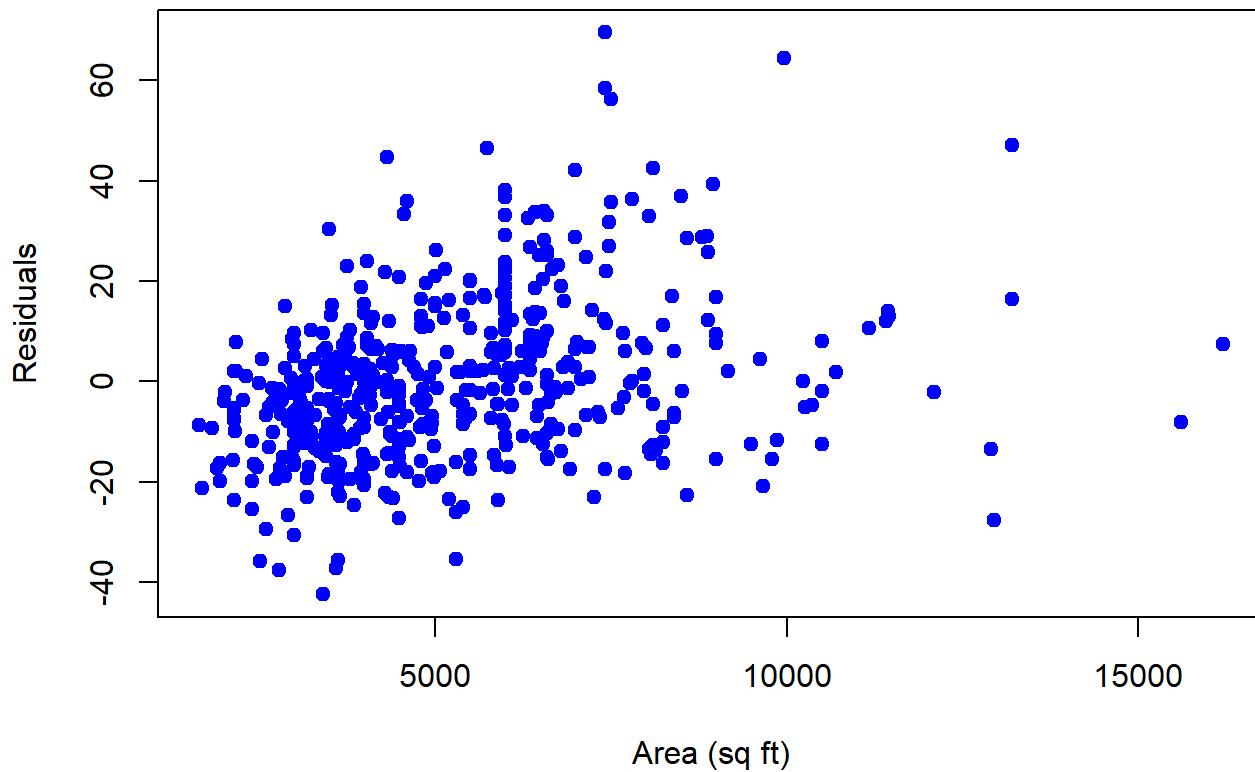
```
# Do the same, but for all variables. Note! This is only to add a experiment with a third model, even though the assignment only calls for 2. For the major of tasks, this will model will be omitted, but we'll use it for the final ANOVA.
```

```
all_r_squared <- summary(all_variables_lm_model)$r.squared
all_adj_r_squared <- summary(all_variables_lm_model)$adj.r.squared
all_rmse <- sqrt(mean(all_variables_lm_model$residuals^2))
all_mse <- mean(all_variables_lm_model$residuals^2)
```

Get the residuals of your second model (you can use 'resid' or 'residuals' functions) and plot them. What does the plot tell you about your predictions?

```
# Get residuals
residuals_lm_multi <- resid(multi_lm_model)
# Plot residuals
plot(
  housing_df$area, residuals_lm_multi,
  main = "Residuals of Linear Regression Model",
  xlab = "Area (sq ft)", ylab = "Residuals",
  pch = 19, col = "blue"
)
```

Residuals of Linear Regression Model



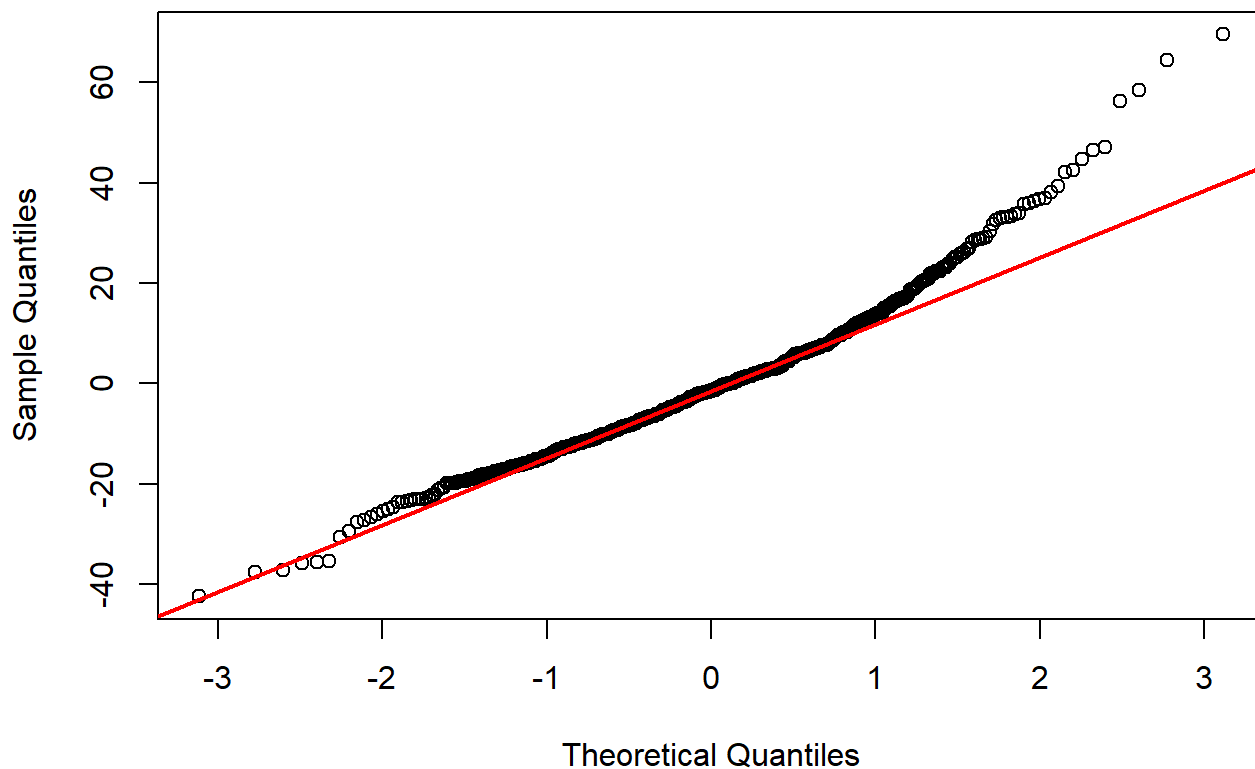
```
cat("The residuals show more variance as area increases. This suggests that the model's  
predictions are less accurate for larger areas, indicating potential non-linearity or the  
need for additional predictors.")
```

```
## The residuals show more variance as area increases. This suggests that the model's pr  
edictions are less accurate for larger areas, indicating potential non-linearity or the  
need for additional predictors.
```

Use a qq plot to observe your residuals. Do your residuals meet the normality assumption?

```
# Q-Q plot of residuals  
qqnorm(residuals_lm_multi)  
qqline(residuals_lm_multi, col = "red", lwd = 2)
```

Normal Q-Q Plot

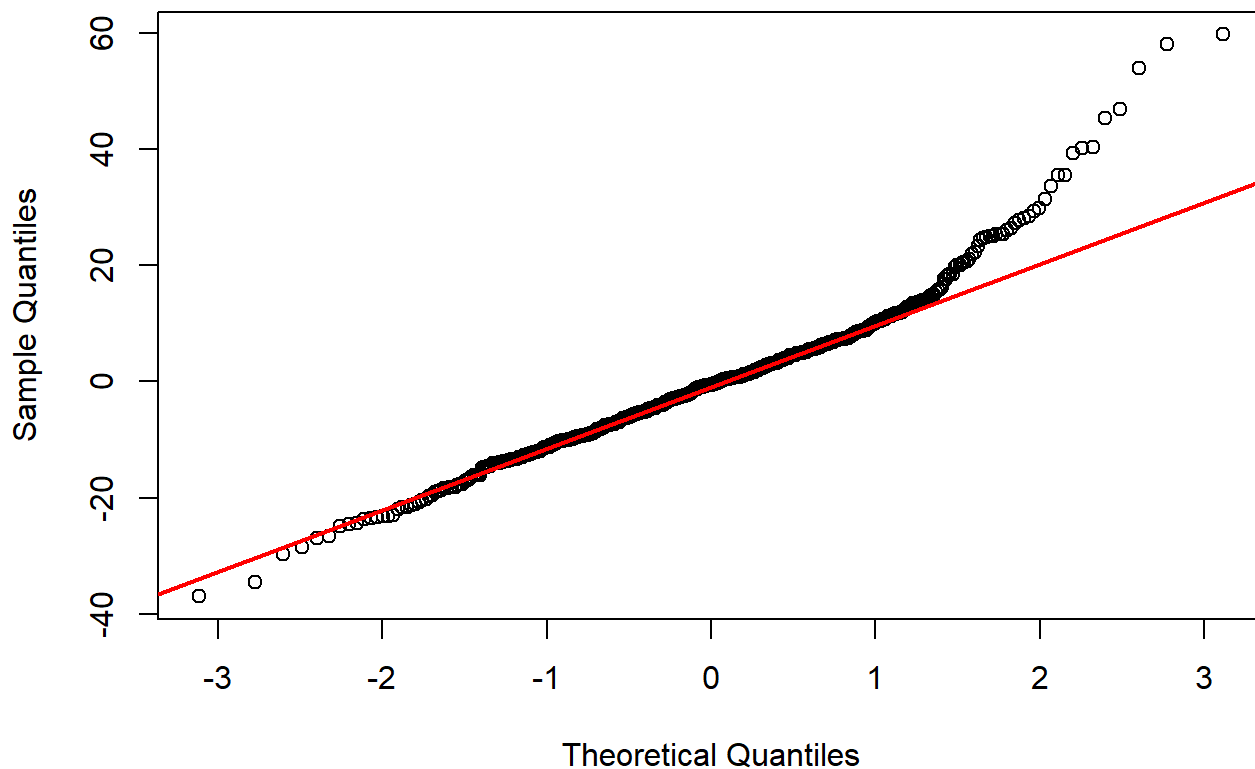


```
cat("Both the end and beginnig of the Q-Q plot show that the residuals do not meet the normality assumption. The residuals deviate from the diagonal line. Similar to the first model, the residuals are not normally distributed, indicating that the assumptions of linear regression may not be fully met.")
```

```
## Both the end and beginnig of the Q-Q plot show that the residuals do not meet the normality assumption. The residuals deviate from the diagonal line. Similar to the first model, the residuals are not normally distributed, indicating that the assumptions of linear regression may not be fully met.
```

```
# Since both failed, I'm going to try using all variables
residuals_all_lm_multi <- resid(all_variables_lm_model)
qqnorm(residuals_all_lm_multi)
qqline(residuals_all_lm_multi, col = "red", lwd = 2)
```

Normal Q-Q Plot



Compare the results (e.g., R2, adj R2, RMSE, MSE) between your first and second model. Does your new model show an improvement over the first? To confirm a 'significant' improvement between the second and first model, use ANOVA to compare them. What are the results?

```
# Compare models using ANOVA, but using 3 models. One contains only area, one contains only new features, the last contains both old and new
anova(lm_model, multi_lm_model, all_variables_lm_model)
```

```
## Analysis of Variance Table
##
## Model 1: price_usd_100k ~ area
## Model 2: price_usd_100k ~ total_rooms + space_per_story
## Model 3: price_usd_100k ~ total_rooms + space_per_story + area + bedrooms +
##          bathrooms + stories
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1     543 135643
## 2     542 131540  1      4103 25.022 7.682e-07 ***
## 3     539  88385  3     43155 87.723 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##      bathrooms + stories
## Res.Df    RSS Df Sum of Sq      F    Pr(>F)
## 1      543 135643
## 2      542 131540  1      4103 25.022 7.682e-07 ***
## 3      539  88385  3     43155 87.723 < 2.2e-16 ***
```

```
cat("The ANOVA comparison shows that each step of adding variables significantly improves the model.\n")
```

```
## The ANOVA comparison shows that each step of adding variables significantly improves the model.
```

```
cat("Model 2, which adds one variable to the base model (bathrooms + stories), significantly reduces the residual sum of squares (RSS) with an F-statistic of 25.02 and a p-value of 7.68e-07.\n")
```

```
## Model 2, which adds one variable to the base model (bathrooms + stories), significantly reduces the residual sum of squares (RSS) with an F-statistic of 25.02 and a p-value of 7.68e-07.
```

```
cat("Model 3 adds three more variables, leading to a much larger drop in RSS and an F-statistic of 87.72 with a p-value less than 2.2e-16.\n")
```

```
## Model 3 adds three more variables, leading to a much larger drop in RSS and an F-statistic of 87.72 with a p-value less than 2.2e-16.
```

```
cat("This indicates that each additional set of predictors provides substantial explanatory power, and Model 3 is statistically the best-fitting model of the three.\n")
```

```
## This indicates that each additional set of predictors provides substantial explanatory power, and Model 3 is statistically the best-fitting model of the three.
```